Analytic Preliminaries

Hölder spaces Let $\Omega \subset \mathbb{R}^n$ be open, $\alpha \in (0,1)$, $k \in \mathbb{N}$.

For $f : \Omega \to \mathbb{R}$ define

$$|f|_{C^\alpha} = \sup_{x \neq y \in \Omega} \frac{|f(x) - f(y)|}{|x-y|^\alpha}$$

Define $$\|f\|_{C^k} = \max_{|\beta| \leq k} \sup_{x \in \Omega} |D^\beta f(x)|$$

$$\beta = (\beta_1, \dots, \beta_n) \qquad |\beta| = \beta_1 + \dots + \beta_n$$

$$D^\beta f = \frac{\partial^{|\beta|} f}{\partial x_1^{\beta_1} \dots \partial x_n^{\beta_n}}$$

$C^{k,\alpha}$ Hölder norm is

$$\|f\|_{C^{k,\alpha}(\Omega)} = \|f\|_{C^k} + \max_{|\beta| = k} |D^\beta f|_{C^\alpha}$$

$$C^{k,\alpha}(\Omega) = \{ f : \Omega \to \mathbb{R} \mid \|f\|_{C^{k,\alpha}} < \infty \}$$

Some properties

- $C^{k,\alpha}(\Omega)$ is a Banach space with norm $\|\cdot\|_{C^{k,\alpha}}$

- multiplicative property:
  if $f, g \in C^{k,\alpha}$ then $fg \in C^{k,\alpha}$
  and $\|fg\|_{C^{k,\alpha}} \leq C_k \|f\|_{C^{k,\alpha}} \|g\|_{C^{k,\alpha}}$
  $C_k > 0$ depend on k.

- If $k+\alpha < \ell+\beta$ and
  $f_1, f_2, \ldots \in C^{\ell,\beta}$ s.t. $\|f_i\|_{C^{\ell,\beta}} < C$
  for uniform C
  then there is a subsequence f_{k_i}
  converging in $C^{k,\alpha}$.

  Simplest case: $k = \ell = 0$
  If $\|f_i\|_{C^\beta} < C$
  $\Rightarrow$ by Arzela - Ascoli thm
  $\exists$ subsequence f_{k_i} converging to f uniformly
  $f \in C^\beta$ since $\dfrac{|f(x)-f(y)|}{|x-y|^\beta} = \lim \dfrac{|f_{k_i}(x) - f_{k_i}(y)|}{|x-y|^\beta} \leq C$

<u>Note</u>: for any u we have

$$\frac{|u(x)-u(y)|}{|x-y|^{\alpha}} = \left(\frac{|u(x)-u(y)|}{|x-y|^{\beta}}\right)^{\alpha/\beta} |u(x)-u(y)|^{\overset{>0}{1-\alpha/\beta}}$$

$$\leq \|u\|_{C^{\beta}}^{\alpha/\beta} \left(2\sup|u|\right)^{1-\alpha/\beta}$$

$f_{k_i} \longrightarrow f$ uniformly, so $\sup|f_{k_i} - f| \to 0$

and $\|f_{k_i} - f\|_{C^{\beta}} < C$

$\Longrightarrow \|f_{k_i} - f\|_{C^{\alpha}} \longrightarrow 0$

i.e. $f_{k_i} \longrightarrow f$ in C^{α}

- Alternative characterization

<u>Theorem</u> Let $\alpha \in (0,1)$. $\exists C > 0$ satisfying the following. Suppose $f : B \to \mathbb{R}$ on ball B and for all $x \in B$ there is a degree 2 poly $P_x(y) = \frac{1}{2} a_{x,ij} y_i y_j + b_{x,i} y_i + c_x$ with $|a_{ij}| + |b_i| + |c| \leq 1$ and

$$\sup_{B(x,r)} |f - P_x| \leq r^{2+\alpha} \quad \forall r < 1 \quad (*)$$

Then $\|f\|_{C^{2,\alpha}(B)} \leq C$.

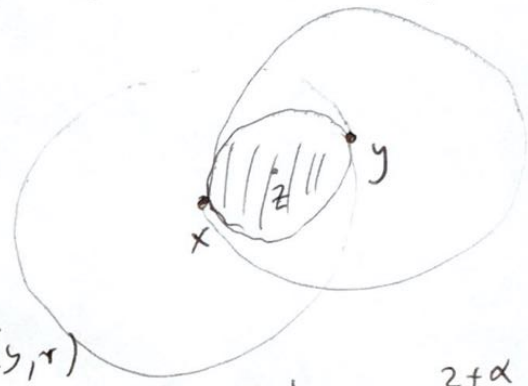
<u>Proof</u> Note that $(*) \Rightarrow f(x) = P_x(x)$

$$D^\beta f(x) = D^\beta P_x(x)$$

for $|\beta| \le 2$.

$$\partial_i \partial_j f(x) = a_{x,ij}$$

To prove $\|f\|_{C^{2,\alpha}} \le C$ we need to show $\dfrac{|a_{x,ij} - a_{y,ij}|}{|x-y|^\alpha} \le C$ for $x \ne y \in B$



let $r = |x-y|$

$z = \dfrac{x+y}{2}$

$(*)$ applied to $B(x,r)$, $B(y,r)$

$$\Rightarrow \sup_{B(x,r)} |f - P_x|, \ \sup_{B(y,r)} |f - P_y| \le r^{2+\alpha}$$

$$\Rightarrow \sup_{B(z,r/2)} |P_x - P_y| \le 2r^{2+\alpha}$$

$$\Rightarrow |a_{x,ij} - a_{y,ij}| \le C r^\alpha$$

so $\dfrac{|a_{x,ij} - a_{y,ij}|}{|x-y|^\alpha} \le C$

- On a compact manifold M define $\|f\|_{C^{k,\alpha}}$ for $f: M \to \mathbb{R}$ by covering M with finitely many charts

$$M = \bigcup_{i=1}^{N} U_i \quad , \quad \varphi_i : U_i \to V_i \subset \mathbb{R}^n$$

$$f \circ \varphi_i^{-1} : V_i \longrightarrow \mathbb{R}$$

Define $\|f\|_{C^{k,\alpha}(M)} = \max_i \|f \circ \varphi_i^{-1}\|_{C^{k,\alpha}(V_i)}$

<u>Note</u>: norm depends on choice of charts
Different charts lead to equivalent norms.

The Laplacian on $\mathbb{R}^n$

$$\Delta f = \frac{\partial^2 f}{\partial x_1^2} + \dots + \frac{\partial^2 f}{\partial x_n^2}$$

Dirichlet problem in a ball: $B = B(0,1) \subset \mathbb{R}^n$

If $g : \partial B \longrightarrow \mathbb{R}$ is continuous
then $\exists\, f : B \longrightarrow \mathbb{R}$ s.t.

$$\begin{cases} \Delta f = 0 & \text{in } B \\ f = g & \text{on } \partial B \end{cases} \qquad (*)$$

Can use Poisson kernel $P(x,y) = \dfrac{1-|x|^2}{\omega_{n-1}\,|x-y|^n}$

$x \in B$, $y \in \partial B$,

ω_{n-1} = surface area of ∂B.

$\Rightarrow$ solution of $(*)$ is

$$f(x) = \int_{\partial B} g(y)\, P(x,y)\, dA_y$$